

MISSION 4 – LIVE GRAPHING (ROLL, PITCH, YAW)

Objective

To evaluate drone flight stability by interpreting live sensor data (roll, pitch, yaw) during the Mission 1 waypoint route and correlating attitude changes with specific flight events.

Axis Definitions

Roll — Side tilt of the drone (positive = right bank, negative = left bank). Returns to zero during straight, level flight.

Pitch — Forward/backward tilt (positive = nose up during acceleration, negative = nose down during braking). Near zero at constant cruise speed.

Yaw — Compass heading direction. Changes gradually during turns, stable on straight legs, spikes sharply at tight waypoint turns.

Timestamped Flight Interpretation

T = 0:14 — Takeoff from Tech Green (Launch)

- Pitch goes slightly positive as the drone accelerates forward toward Bobby Dodd.
- Roll near zero; yaw stable at initial heading.
- Confirms clean takeoff with no lateral drift.

T = 0:20 — Turning toward Bobby Dodd Stadium (WP1)

- Roll spikes as the drone banks into the turn toward Bobby Dodd.
- Yaw begins changing, reflecting the new heading.
- Once the turn completes, roll returns near 0, confirming PID re-stabilization.

T = 0:40 — Turning toward Couch Park (WP2)

- Roll spikes as the drone banks into the turn.
- Yaw shifts steadily as heading changes toward Couch Park.
- Pitch briefly dips negative as speed reduces entering the waypoint.
- Roll recovers near 0 as the drone exits the turn and straightens.

T = 0:42 — Straight leg, Bobby Dodd → Couch Park (WP2)

- All three axes stabilize: roll $\approx 0^\circ$, pitch $\approx$ small positive constant, yaw fixed on heading.
- Small oscillations ($\pm 1-2^\circ$) in roll and pitch reflect healthy PID correction, not instability.

T = 1:25 — Turning toward CRC (WP3)

- Roll spikes in the direction of the bank.
- Yaw changes as heading adjusts toward CRC.
- Roll and pitch return to baseline after turn completion.

T = 1:26 — Straight leg, Couch Park → CRC (WP3)

- Roll holds near 0° , yaw fixed, pitch slightly positive at cruise.
- Stable segment with minimal corrections.

T = 1:39 — Turning toward Van Leer (WP4)

- Roll spikes with banking motion.
- Yaw shifts to new heading toward Van Leer.
- Recovery to stable attitude within a few seconds.

T = 1:40 — Straight leg, CRC → Van Leer (WP4)

- All axes stabilize at cruise values.
- Demonstrates baseline stable flight between waypoints.

T = 2:04 — Turning toward Coliseum (WP5)

- Depending on heading change sharpness, roll spike may be one of the larger ones in the mission.
- Yaw changes rapidly; pitch dips slightly negative through the turn.
- PID recovery visible within seconds of exiting the turn.

T = 2:05 — Straight leg, Van Leer → Coliseum (WP5)

- Roll $\approx 0^\circ$, pitch and yaw stable.
- Continued stable cruise behavior observed.

T = 2:33 — Turning toward Russ Chandler (WP6)

- Roll spikes with banking into the final waypoint turn.
- Yaw adjusts to heading toward Russ Chandler.
- Roll returns near 0 on straightening.

T = 2:34 — Straight leg, Coliseum → Russ Chandler (WP6)

- Final cruise leg before RTL.
- All axes stable; roll near 0, yaw fixed on heading to Russ Chandler.

T = 2:49 — RTL Initiated, returning to Tech Green

- Yaw shifts back toward launch heading — visible as a smooth, continuous change.
- Roll oscillates slightly during the heading correction, then stabilizes.
- Pitch goes slightly negative as the drone descends back to 50 ft before landing.

Stability Analysis

- Small oscillations of $\pm 1\text{--}3^\circ$ in roll and pitch throughout the mission indicate healthy, active PID control — not instability.
- No continuous drift was observed on any axis; all deviations returned to baseline after every turn.
- Roll spikes correlated directly and consistently with every waypoint turn across all 6 waypoints, and returned near zero on every straight leg — confirming correct banking and recovery behavior.
- Yaw changes were smooth and proportional to turn sharpness at each waypoint.
- Pitch behavior correctly reflected acceleration and braking events at each waypoint transition.

Conclusion

The drone maintained stable flight dynamics throughout all 6 waypoints of the Mission 1 route. Live roll, pitch, and yaw data confirmed that attitude changes were logically tied to flight events: banking during turns, pitch changes during acceleration and braking, and yaw tracking heading transitions at every waypoint from Bobby Dodd through Russ Chandler.

The absence of continuous drift and the consistent return-to-zero behavior on every straight leg confirms that the PID controller maintained effective attitude control for the full duration of the mission and RTL.